

Network Cable and connectors

- 1) Draw a labeled diagram of a Cat6 Ethernet cable, showing the color order of wires in the T568B standard and labeling the RJ-45 connector pins.

Pin	Wire Color
1	White/Orange
2	Orange
3	White/Green
4	Blue
5	White/Blue
6	Green
7	White/Brown
8	Brown

- 2) List three types of network cables (such as Cat5e, Cat6, and fiber optic) and for each, mention one real-world app or device (like a WiFi router, Zomato POS terminal, or IPL streaming server) where it would be used, explaining why.

Cable Type	Real-World Device/Application	Why It Is Used
Cat5e	Wi-Fi Router	Supports Gigabit Ethernet and is affordable for homes.
Cat6	Zomato POS Terminal	Provides faster speed (up to 1 Gbps over 100 m, higher over shorter distances) with less interference.
Fiber Optic	IPL Streaming Server	Very high speed, long-distance communication, and low signal loss for live video streaming.

3) Watch a YouTube video showing how to crimp an RJ-45 connector onto a network cable, then write down the step-by-step process in your own words and mention two common mistakes to avoid.

1. Cut the Ethernet cable to the required length.
2. Strip about 2–3 cm of the outer jacket.
3. Untwist the wire pairs carefully.
4. Arrange the wires in the T568B color order.
5. Trim all wires evenly.
6. Insert the wires fully into the RJ-45 connector.
7. Ensure each wire reaches the end of the connector.
8. Place the connector into the crimping tool.
9. Squeeze the crimping tool firmly.
10. Test the cable using a network cable tester.

○ **Two Common Mistakes**

- Mixing up the wire color order.
- Not pushing the wires completely to the end of the RJ-45 connector before crimping.

4) Given the following scenario: You are setting up a gaming zone for an IPL screening event and need to connect multiple screens and consoles. Choose the appropriate cable and connector types for high-speed, reliable connections, and explain your choices.

- **Cable**

- Cat6 Ethernet Cable

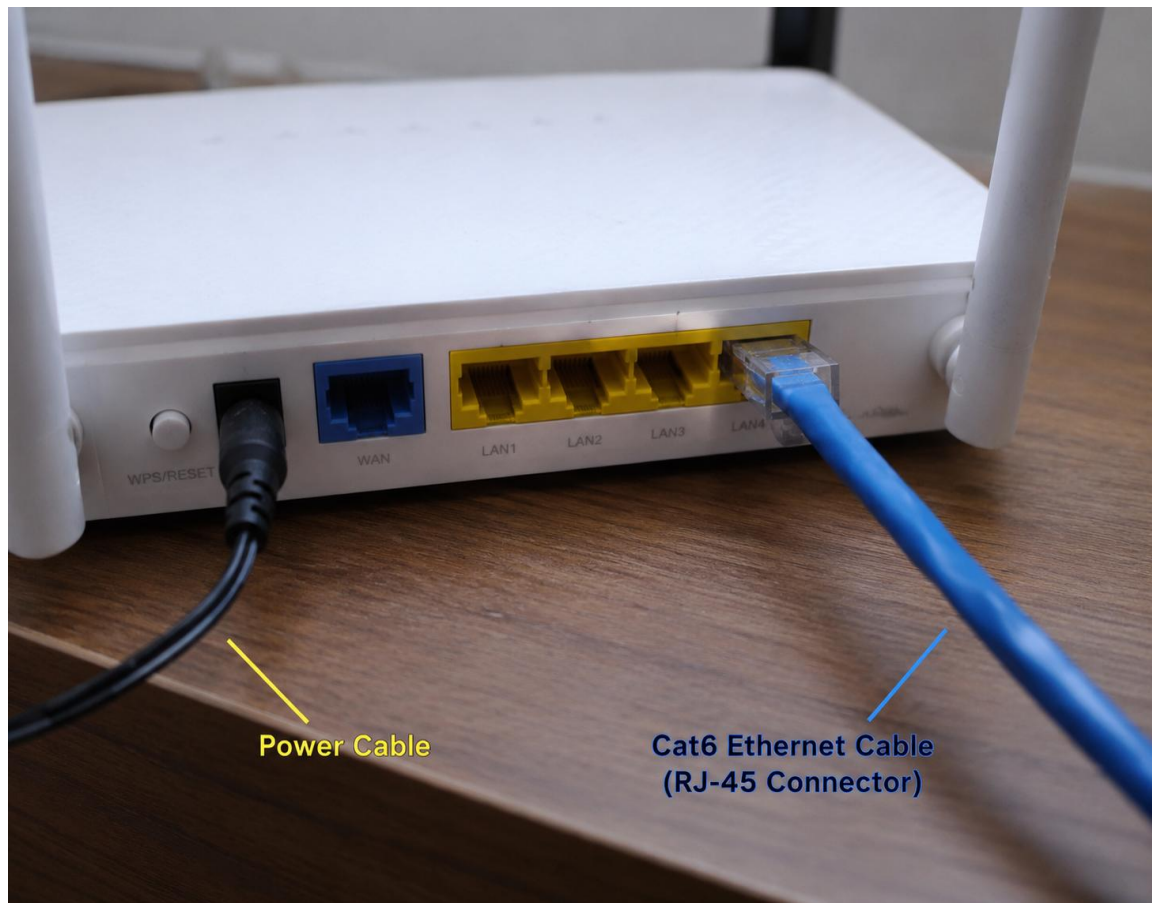
- **Connector**

- RJ-45 Connector

- Supports high-speed network connections.
- Low latency for online gaming.
- Reliable and stable compared to Wi-Fi.
- Handles multiple gaming consoles and smart TVs simultaneously.
- Cost-effective and easy to install.

- 5) Find and share a photo (from your home, college, or online) of any network cable or connector in use (such as behind a WiFi router or at a café's POS counter), and describe what type it is and what devices it connects.

Hint: Look for blue or yellow cables with clear plastic ends (RJ-45) or thin round cables for fiber optics.



This is a **Cat6 Ethernet cable** with an **RJ-45 connector**. It is connected between the Wi-Fi router and a computer. It provides a stable, high-speed internet connection for browsing, online gaming, video streaming, and file transfers. The RJ-45 connector locks securely into the Ethernet port to prevent accidental disconnection.